PampaR v9: Brain-Inspired Territorial Architecture

A Language Model with Interpretable Routing

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Abstract

PampaR is a language model architecture inspired by the functional organization of the human brain. Unlike standard transformers that treat all computations uniformly, PampaR introduces **Territorial Processing** where specialized neural modules are organized into functional territories coordinated by a central **Thalamus** that routes tokens using a hybrid approach: 70% explicit rules (LLAVES) and 30% learned attention.

> **Key Result:** ~45 perplexity on WikiText-103 with only 14M parameters, outperforming LSTM (24M, PPL 69.1) and approaching Transformer-XL Small (24M, PPL 54.5) with 42% fewer parameters.

Architecture Overview

```
Input → Embedding → [Territorial Block ×N] → LM Head → Output
                          |
                 ┌────────────────────┐
                 │       TÁLAMO       │
                 │ LLAVES 70% + Attn 30% │
                 └────────────────────┘
                          |
        ┌─────────────────┼─────────────────┐
        ▼                 ▼                 ▼
┌──────────────┐   ┌──────────────┐   ┌──────────────┐
│  EXPRESIVO   │◄─►│  CONTEXTUAL  │◄─►│    FORMAL    │
│Language+Creat.│  │   Context    │   │    Logic     │
└──────────────┘   └──────────────┘   └──────────────┘
       |                  |                  |
       |           ┌──────────────┐          |
       └───────────│ ESTRUCTURAL  │──────────┘
                   │ Pattern + Math │
                   └──────────────┘
```

Territories

PampaR organizes computation into 4 functional territories, each containing specialized modules:

Territory	Modules	Function	Brain Analogy

Expresivo	Language, Creativity	Fluent text generation, novel ideas	Broca's area, Right hemisphere
Contextual	Context	Working memory, coherence	Hippocampus, Prefrontal cortex
Formal	Logic	Deductive reasoning, rules	Prefrontal cortex
Estructural	Patterns, Mathematics	Sequences, numbers, structure	Parietal lobe

🔑 LLAVES: Interpretable Routing

LLAVES (Spanish for "keys") are explicit, interpretable routing rules based on token patterns:

Module	Activation Patterns
Language	"the", "a", "of", "that", articles, prepositions
Mathematics	digits 0-9, "+", "-", "=", "%", operators
Logic	"if", "then", "because", "therefore"
Patterns	repetitions, sequences, regular structures
Context	pronouns, references, temporal markers
Creativity	adjectives, metaphors, novel combinations

The routing formula combines rules with learning:

> `routing_weights = λ × LLAVES_activation + (1 - λ) × learned_attention` where λ = 0.7

⇌ Bidirectional Frontiers

Territories communicate via 6 bidirectional frontier connections with learned gates:

Connection	Weight	Function
Expresivo ↔ Contextual	0.8 (High)	Narrative needs context
Expresivo ↔ Formal	0.5 (Medium)	Structured argumentation
Expresivo ↔ Estructural	0.4 (Low)	Creative structure
Contextual ↔ Formal	0.6 (Medium-high)	Logical context
Contextual ↔ Estructural	0.5 (Medium)	Patterns in context
Formal ↔ Estructural	0.7 (High)	Mathematical logic

⚙ Model Configuration

Parameter	Value
Total Parameters	14M
Hidden Dimension	256
Attention Heads	8
Territorial Blocks	6
Context Length	512 tokens
Vocabulary	32K (BPE)

Training	Value
Dataset	WikiText-103
Hardware	GTX 1650 (4GB)
Optimizer	AdamW
Learning Rate	3e-4
Precision	FP16 mixed
Training Time	~8 hours

Results Comparison

Model	Parameters	PPL ↓	Year
LSTM (Merity et al.)	24M	69.1	2018
Transformer-XL Small	24M	54.5	2019
PampaR v9	**14M**	**~45**	2026
GPT-2 Small	125M	35.1	2019

Key Advantages

- **Interpretability:** LLAVES provide explicit routing rationale
- **Parameter Efficiency:** Competitive PPL with 42% fewer parameters
- **Modularity:** Easy to add or modify specialized territories
- **Biological Plausibility:** Reflects brain's functional organization

Limitations & Future Work

- Scale: Not yet tested at 1B+ parameters
- Tasks: Evaluated mainly on language modeling; needs downstream evaluation
- LLAVES Design: Currently manual; could benefit from automatic rule learning

Reproducibility

GitHub: `github.com/lucasmella-stack/PAMPAr-o1`

HuggingFace: `huggingface.co/lucasmella-stack/PAMPAr-o1`

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